

MALITHA GUNAWARDHANA

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SUMMARY

Machine Learning and Software Engineer with over four years of experience developing practical AI systems across computer vision, medical imaging, anomaly detection, video analytics, seismic signal processing, sensor data, and predictive modelling. Proficient in Python, PyTorch, TensorFlow, Keras, and Scikit-learn, with experience in data engineering, model evaluation, signal processing, and scalable software development. Currently developing time-series analysis and anomaly-detection solutions at Viotel, working with real-world accelerometer and seismic sensor data. Experienced in collaborating with engineers, researchers, clinicians, and operational stakeholders to translate complex problems into reliable, responsible, and measurable solutions. Completing advanced doctoral research at the University of Auckland, with thesis submission expected in August 2026.

EXPERIENCE

- Data Processing Technician, Part-Time | Machine Learning and Signal Processing** March 2026 – Present
Viotel Pty Ltd *Parnell, Auckland*
- Developed machine learning-based anomaly detection workflows for accelerometer data collected from **more than eight telecommunications towers**, supporting the automated identification of abnormal structural vibrations.
 - Trained and evaluated deep learning models for automated P- and S-wave phase picking using **3,729 seismic events and more than 30,000 station samples**, with the aim of substantially reducing manual waveform-picking time.
 - Collaborated with a three-member field team to install **17 seismic sensors** across the Taupō region, enabling successful deployment and continuous seismic data collection.
- Machine Learning Engineer** Sep. 2022 – Jan. 2025
Institute of Fundamental Technological Research, Polish Academy of Sciences (IPPT PAN) *Warsaw, Poland*
- Led the development and refinement of deep learning models for tumour detection and segmentation, working closely with clinical collaborators to improve model performance.
 - Designed and implemented scalable artificial intelligence pipelines for medical image analysis, data processing, model training, evaluation, and deployment.
 - Collaborated with more than three cross-functional teams, including clinicians, researchers, and engineers, to align artificial intelligence solutions with clinical and research requirements.
- Artificial Intelligence Researcher** Sep. 2022 – Sep. 2023
Mohamed bin Zayed University of Artificial Intelligence (MBZUAI) *Abu Dhabi, UAE*
- Designed and implemented artificial intelligence frameworks for perceptual understanding, contributing to the development of robust self-supervised and semi-supervised learning techniques.
 - Conducted research in network calibration, large language models, video analysis, self-supervised learning, and semi-supervised learning.
 - Led and collaborated with a four-member research team to produce high-quality research and contribute to publications in leading artificial intelligence conferences.
- Machine Learning Engineer** June 2022 – Nov. 2022
PromiseQ GmbH *Berlin, Germany*
- Improved the accuracy and reliability of a deep learning-based CCTV surveillance system designed to detect activities in real time by integrating advanced object detection and anomaly recognition algorithms.
 - Reduced false alarms by **5%** using network calibration techniques, improving overall system reliability and user trust.
 - Designed scalable artificial intelligence pipelines for efficiently processing video streams while balancing computational cost and detection accuracy.
 - Collaborated with cross-functional teams to deploy artificial intelligence-driven surveillance solutions supporting security and operational monitoring.
- Full-Stack Software Engineer** March 2021 – May 2022
Xeptagon (Pvt) Ltd *Colombo, Sri Lanka*
- Led the development of a **domain drop-catching platform** using optimised algorithms and multithreaded backend engineering, achieving a success rate above **90%** with latency below **50 milliseconds**.
 - Designed and implemented scalable and responsive backend systems using an iterative feedback process with stakeholders to ensure reliable system performance.
 - Performed feature extraction on more than **80 audio signals** using signal processing and feature engineering techniques to support a student learning management system.
- Software Engineer Intern** June 2019 – Dec. 2019
Synergen Technology Labs (Pvt) Ltd *Colombo, Sri Lanka*
- Designed a wearable device for physiological signal acquisition and developed feature extraction pipelines to calculate numerical stress indicators.
 - Trained a machine learning model to classify stress into relaxed, cognitive stress, physical stress, and emotional stress categories, achieving more than **90% classification accuracy**.

EDUCATION

Doctor of Philosophy

Dec. 2023 – Expected Aug. 2026

University of Auckland

Auckland, New Zealand

- **Thesis:** Deep-learning-powered analysis to support structure-targeted therapy for atrial fibrillation.
- **Scholarship:** Health Research Council Scholarship.
- Conducting deep learning research aimed at improving treatment outcomes for patients with atrial fibrillation.
- Investigating effective fusion mechanisms for pretrained and custom machine learning models.
- Developing efficient data pipelines for machine learning experiments, model training, and evaluation.
- Publishing and presenting research findings in internationally recognised journals and conferences.

B.Sc. Engineering Honours

Jan. 2017 – July 2021

University of Moratuwa

Moratuwa, Sri Lanka

- Achieved Dean's List recognition in Semester 7.
- *Relevant coursework:* Image Processing and Machine Vision, Neural Networks and Fuzzy Logic, Calculus, Linear Algebra, Differential Equations, Statistics, Graph Theory, Medical Imaging, and Signal Processing.

SELECTED CERTIFICATIONS

Oracle Cloud Infrastructure 2024 Generative AI Certified Professional

- Large language model fundamentals, model pretraining, fine-tuning, alignment, generative artificial intelligence, and machine learning operations.

Large Language Model Agents, UC Berkeley

- Large language model agents, autonomous artificial intelligence systems, multi-agent coordination, prompt engineering, retrieval-augmented generation, model deployment, and machine learning operations.

Deep Learning Specialisation, DeepLearning.AI

- Neural networks, convolutional neural networks, recurrent neural networks, deep learning, and sequence modelling.

Data Science Career Track, 365 Data Science

- Python, SQL, data wrangling, data analysis, statistical analysis, machine learning, and model evaluation.

TECHNICAL SKILLS

Programming Languages: Python, Go, MATLAB

Machine Learning: PyTorch, TensorFlow, Keras, Scikit-learn

Data and Distributed Computing: PySpark, Apache Kafka, Apache Airflow

Generative AI and LLMs: LangChain, CrewAI, Weaviate, retrieval-augmented generation, LLM agents

Technical Domains: Deep learning, computer vision, medical imaging, video analytics, anomaly detection, time-series analysis, signal processing, predictive modelling, data science, and analytics

SERVICE AND LEADERSHIP

University of Auckland

2023 – Present

- Elected Postgraduate Student Council Member representing postgraduate students in university committee meetings and contributing to student welfare initiatives and policy discussions.

IEEE EMBS Student Branch Chapter, University of Moratuwa

2018 – 2021

- Served as an advisor for ISC 2021 Moratuwa, an IEEE EMBS conference, supporting event planning and delivery.
- Served as a Council Member from 2020 to 2021 and Treasurer from 2018 to 2019, supporting chapter administration, finances, and operations.

Rotaract Club of the University of Moratuwa and Alumni of the University of Moratuwa

2016 – 2025

- Served as Vice President for Club Service from 2022 to 2023 and Club Service Director from 2021 to 2022.
- Led club service and fellowship activities and received a Bronze Award for “Fellowship under a Specific Audience” at the 2022 Rotaract District Assembly.
- Received the Spirit of Service Award in 2017, 2018, and 2020.